Computer Networks

Subject Code: CD31233

Course Teacher: Dr. Ratnmala N. Bhimanpallewar

Teaching Scheme:

Credits :	3
Lectures / Week :	2Hrs/week
Practical / week :	2 Hrs/ week

Examination Scheme:

CIE:	20
SCE :	20
ESE :	40
OR :	20
Total:	100

Course Objectives:

- To study the fundamentals of networking
- To understand functionalities of Physical and Data link layer
- To understand the functionalities of Network Layer
- To study various protocols at Transport and Application Layer
- To learn different techniques for routing and routing configuration.
- To learn and demonstrate VLAN, ACL and NAT in networking

Introduction:

- **What is computer Network?**

A communication system for connecting computers/hosts.

• A computer network is a number of computers (also known as nodes) connected by some communication lines.

• Two computers connected to the network can communicate with each other through the other nodes if they are not directly connected.

• Some of the nodes in the network may not be computers at all but they are network devices (Like switches, routers etc.) to facilitate the communication.

Introduction:

Uses of the computer Network

• Exchange of information between different computers. (File sharing)

• Interconnected small computers in place of large computers.

• Communication tools (voice , video)

• Some applications and technologies are examples of Distributed system. (Railway reservation system, Distributed databases etc).

Advantages of Computer Network?

• Better communication

• Better connectivity

• Better sharing of Resources

• Bring people together

The Basic computer Network & communication

https://www.youtube.com/watch?v=tj7f244tubM

Unit-I Explore the Network

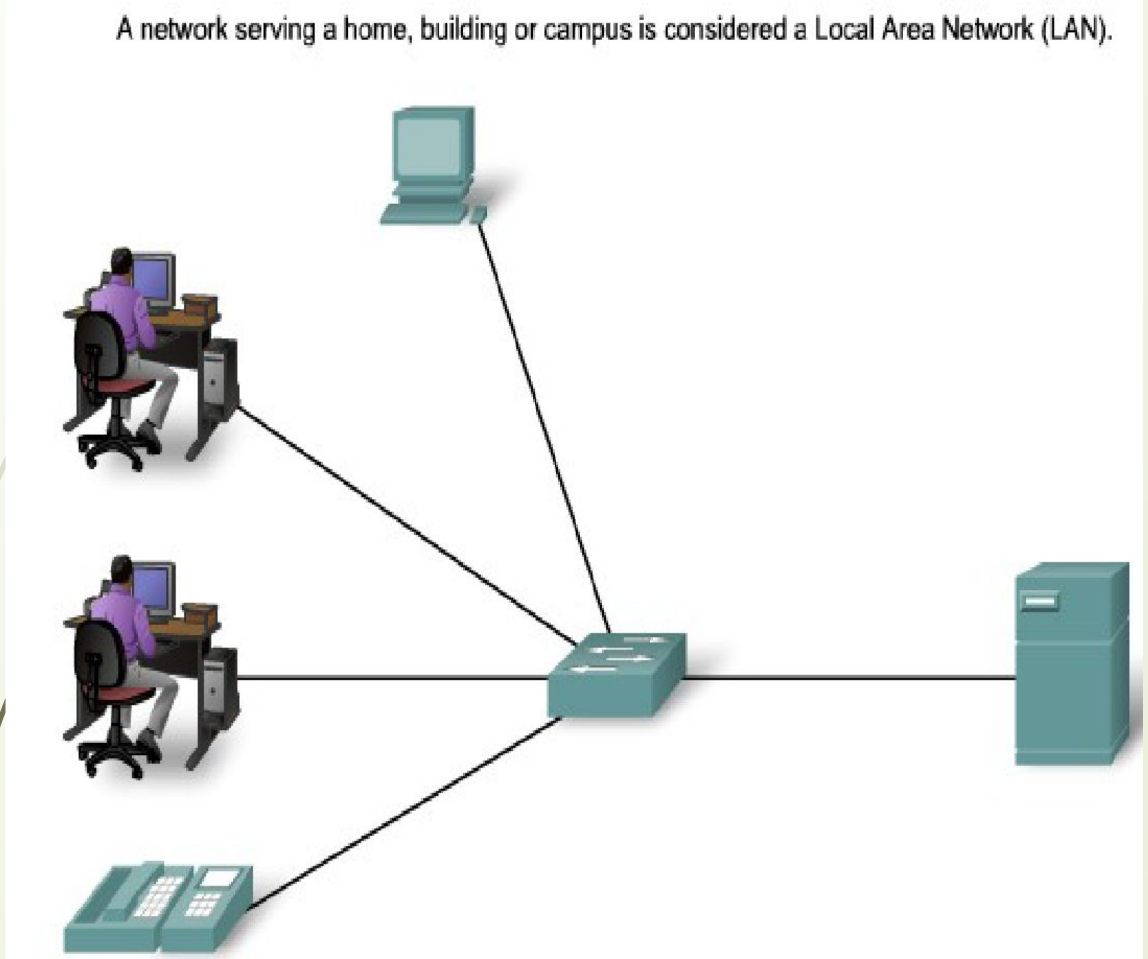
Syllabus:

LANs, WANs, and the Internet, The Network as a Platform, Network Components, Network connecting devices, IEEE standards. Addressing: Physical &logical Addresses, Port Addresses, Specific Addresses.

Rules of Communication: Communication Fundamentals, Rule Establishment, Message Encoding, Message Formatting and Encapsulation, Message Size, Message Timing, Message Delivery options.

LANs, MANS, WANs, and the Internet

LAN



Local Area Network (LAN)

The term LAN refers to a local area network or a group of interconnected network that are under the same administrative control.

In the early days of networking, LANS are defined as small networks that existed in a single physical location. While LANs can be a **single network** installed in a home or small office, the definition of LAN has evolved to include interconnected local networks **consisting of many hundreds of hosts,** installed in **multiple buildings** and locations.

LANs are designed to:

Operate within a limited geographic area.

Allow Multi-access to high bandwidth media.

Can serve thousand users

Can serve upto 5 kilometres, single physical location

LANs

LANs **consist** of the following components:

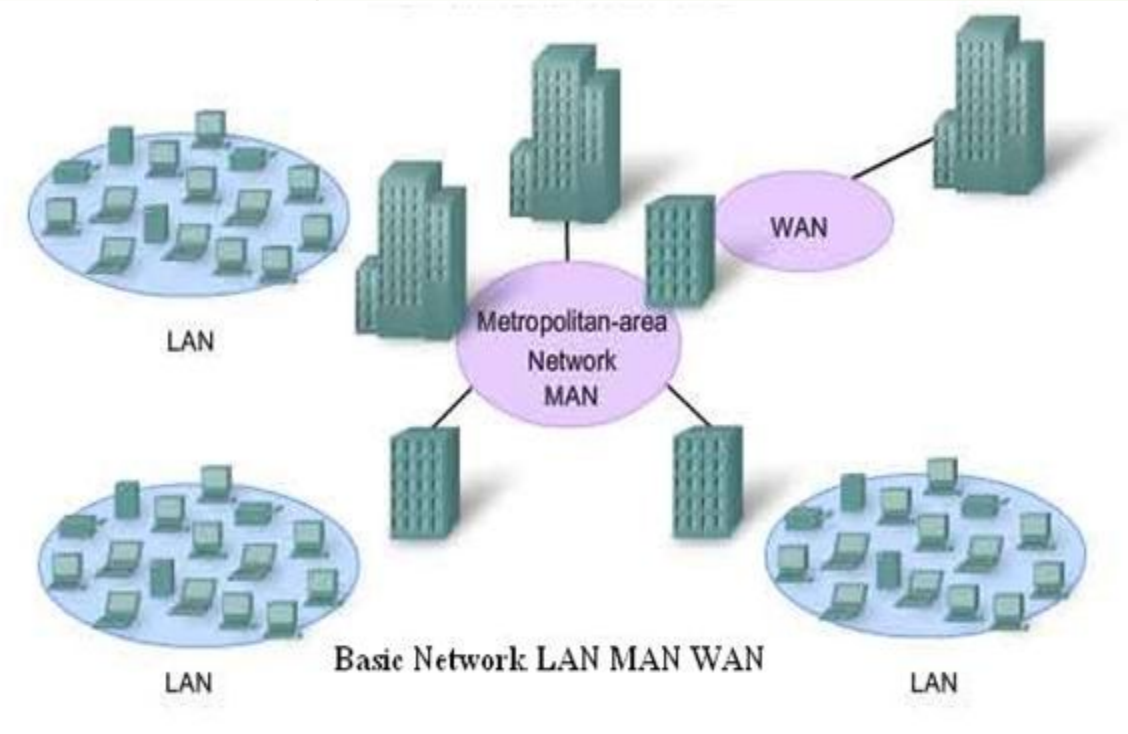
- Computers
- Network interface cards
- Peripheral devices
- Networking media
- Network devices

LANs allow businesses to locally **share computer files and printers efficiently and make internal communications possible.** A good example of this technology is email.

LANs manage data, local communications, and computing equipment. Some common LAN technologies include the following: **Ex**

- Ethernet
- Token Ring
- FDDI- Fiber Distributed Data Interface

MAN



Metropolitan Area Network (MAN)

A MAN, or Metropolitan Area Network, is a computer network that spans a metropolitan area, such as a city or a large campus, connecting multiple LANs. It's larger than a LAN (Local Area Network) but smaller than a WAN (Wide Area Network). MANs are designed to provide high-speed data communication services, often using technologies like fiber optic cables, to connect various locations within a city or region. .

LANs are designed to:

Cover a geographic area of a city or a large campus

Connecting multiple LANs

Facilitate high-speed data transfer

Potentially acting as a bridge to a WAN.

Ex. Uses Fiber optics, Ethernet, Wi-Fi or WiMAX

LANs, WANs, and the Internet :

WAN (Wide Area Network) :

A network that spans **broader geographical area** than a LAN & MAN over **public** communication network.

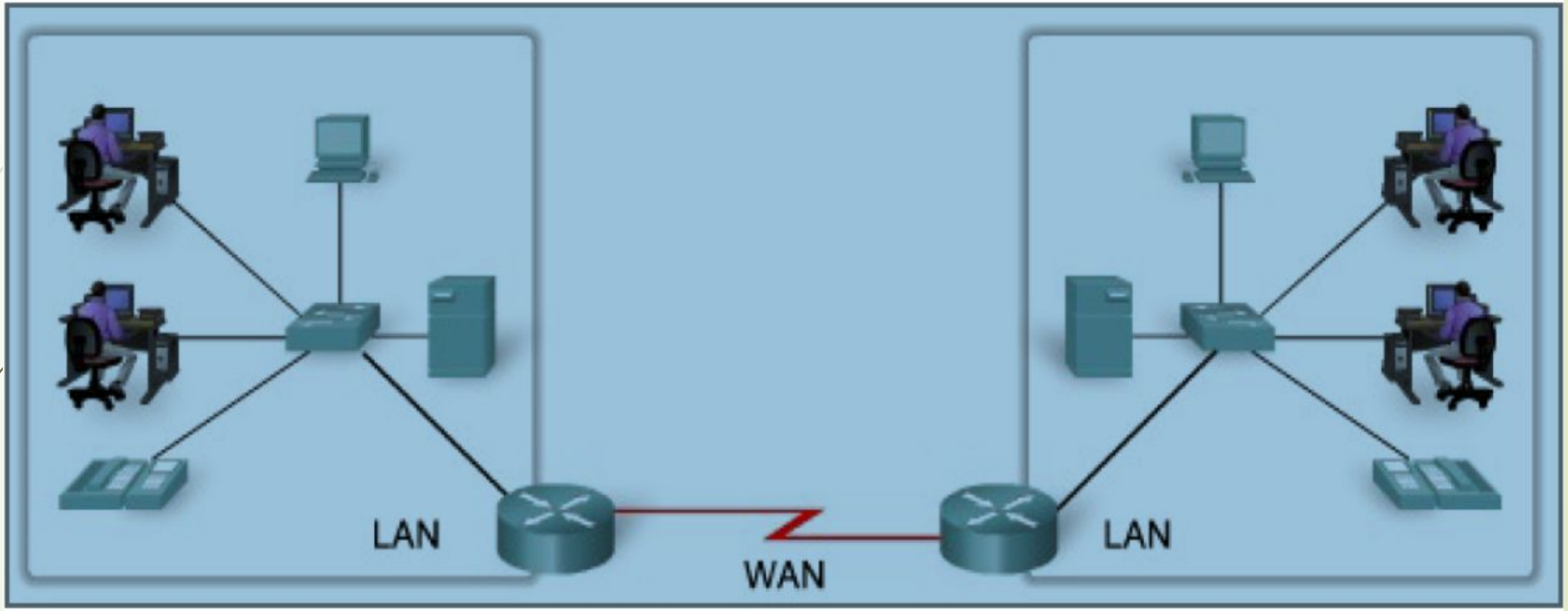
WANs interconnect LANs, which then provide access to computers or file servers in other locations. Because WANs connect user networks over a large geographical area, they make it possible for businesses to communicate across great distances. **WANs allow computers, printers, and other devices on a LAN to be shared with distant locations.** WANs provide instant communications across large geographic areas. Collaboration software provides access to real-time information and resources and allows meetings to be held remotely. WANs have created a new class of workers called telecommuters. **These people never have to leave their homes to go to work.**

WANs are designed to do the following:

- Operate over a large and geographically separated area
- Allow users to have real-time communication capabilities with other users
- Provide full-time remote resources connected to local services
- Provide e-mail, Internet, file transfer, and e-commerce services

Some common WAN technologies include the following:

- Modems
- Integrated Services Digital Network (ISDN)
- Digital subscriber line (DSL)
- Frame Relay
- T1, E1, T3, and E3
- Synchronous Optical Network (SONET)


LAN
WAN
LAN

Internet:

The network formed by the co-operative interconnection of a large number of computer networks.

- Network of Networks
- No one owns the Internet
- Every person who makes a connection owns a slice of the Internet.
- There is no central administration of the Internet.

Internet is comprises of :

A community of people : who use and develop the network.

A collection of resources : that can be reached from those networks.

A setup to facilitate collaboration: Among the members of the research and educational communities world wide.

The connected networks use the TCP/IP protocols:

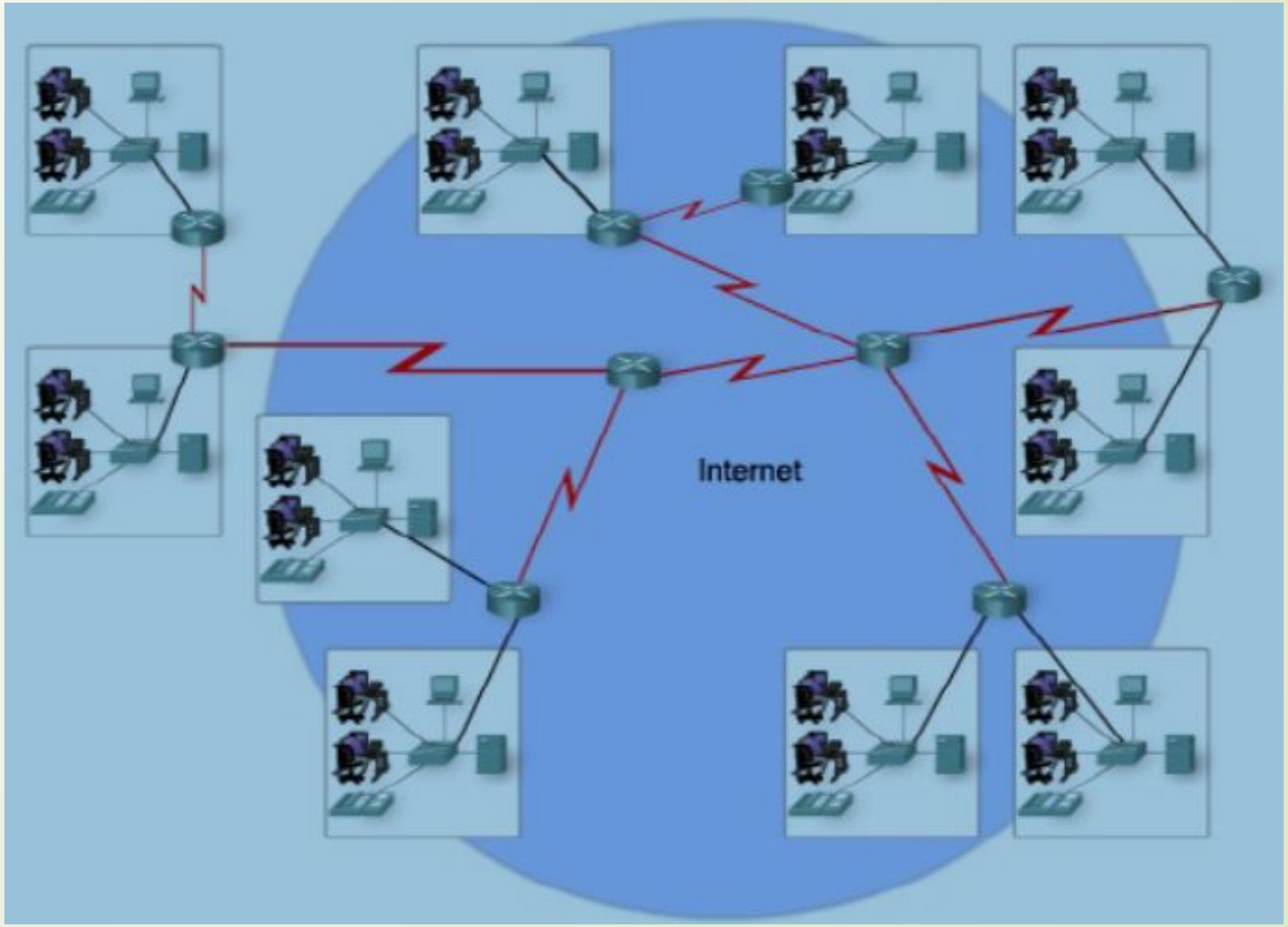
Important Internet applications:

world wide web(WWW)

File Transfer Protocol(FTP)

Electronic Mail

Internet Relay Chat

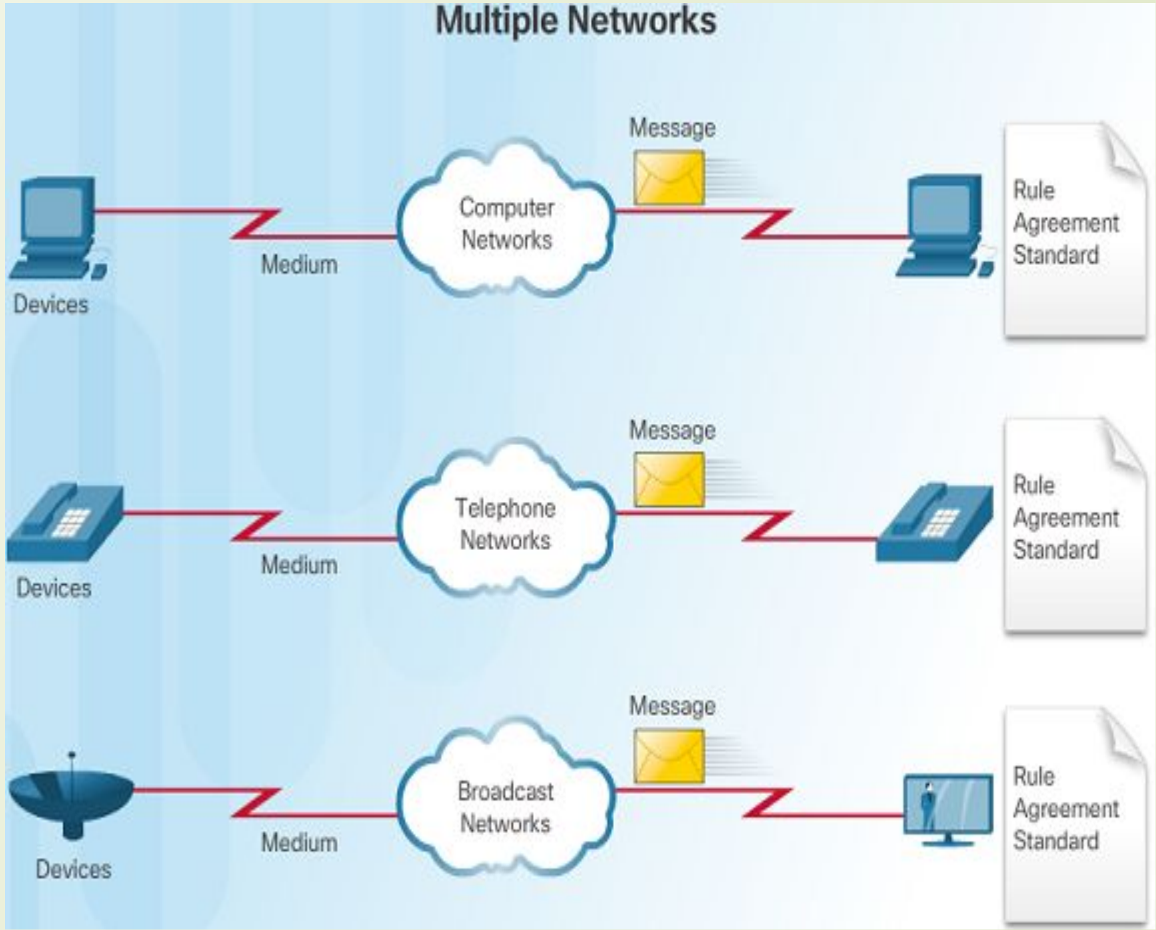


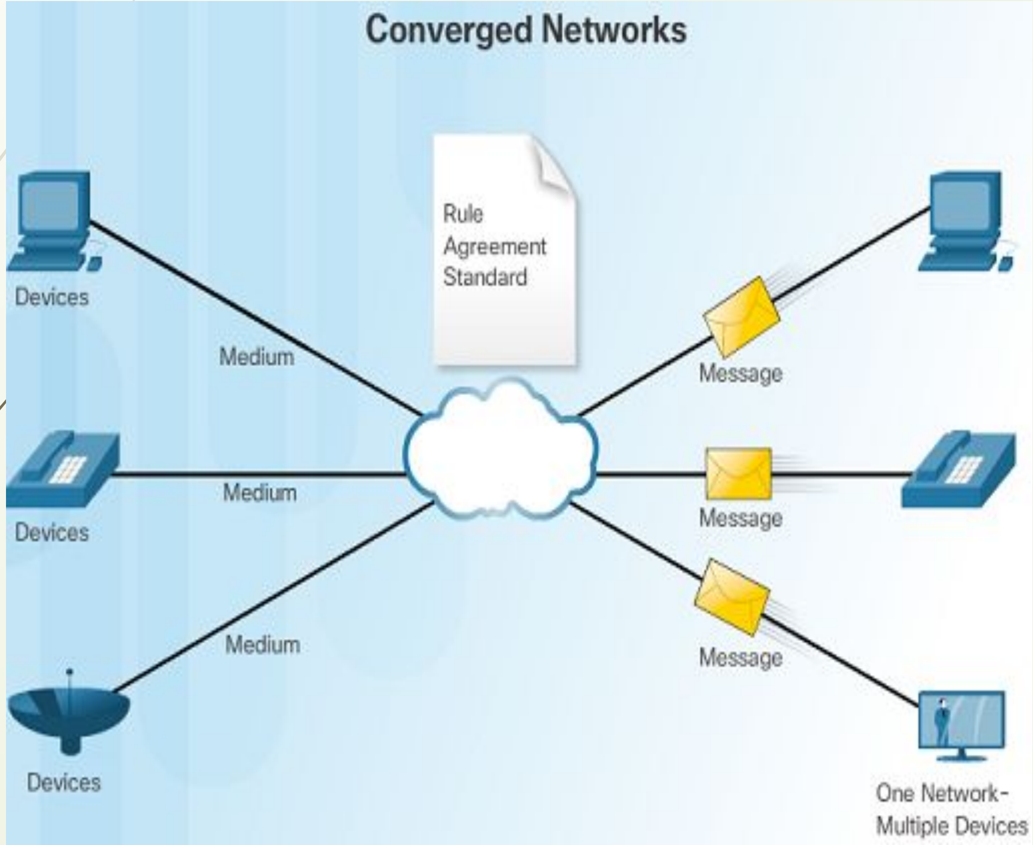
The Network as a Platform

Traditional Separate Networks :

An example of multiple networks might be a school 30 years ago. Some classrooms were cabled for data networks. **Those same classrooms were cabled for telephone networks, and also cabled separately for video**.

Each of these networks used different technologies to carry the communication signals using a different set of rules and standards.





The Converging Network:

Converged data networks carry **multiple services on one link including data, voice, and video.**

Unlike dedicated networks, converged networks can deliver data, voice, and video between different types of devices over the same network infrastructure.

The network infrastructure uses the same set of rules and standards.

Network connecting devices

Network is interconnection of devices.

For these connection we need to use the connecting devices.

Also called as **Network Control Devices**.

- Connectors
- Repeaters
- Hubs
- Bridges
- Switches
- Routers
- NIC’s